Topic: Fourier series representations

Question: Find the Fourier series representation of the function on $-L \leq x \leq L$.

$$f(x) = 1 + x + x^2$$

Answer choices:

A $$f(x) = 1 + \frac{L^2}{3} + \frac{4L^2}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \cos\left(\frac{n\pi x}{L}\right) + \frac{2L}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \sin\left(\frac{n\pi x}{L}\right)$$

B $$f(x) = \frac{L^2}{3} + \frac{4L^2}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \cos\left(\frac{n\pi x}{L}\right) + \frac{2L}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \sin\left(\frac{n\pi x}{L}\right)$$

C $$f(x) = 1 + \frac{L^2}{3} + \frac{4L^2}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \cos\left(\frac{n\pi x}{L}\right)$$

D $$f(x) = \frac{L^2}{3} + \frac{4L^2}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \cos\left(\frac{n\pi x}{L}\right)$$

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Solution: A

The function isn't even because $f(-x) \neq f(x)$ and it's not odd because $f(-x) \neq -f(x)$.

$$f(-x) = 1 - x + (-x)^2$$

$$f(-x) = 1 - x + x^2$$

So we'll start by finding A_0 and the coefficients A_n and B_n. For A_0, we get

$$A_0 = \frac{1}{2L}\int_{-L}^{L} f(x)\, dx$$

$$A_0 = \frac{1}{2L}\int_{-L}^{L} 1 + x + x^2\, dx$$

$$A_0 = \frac{1}{2L}\left(x + \frac{1}{2}x^2 + \frac{1}{3}x^3\right)\Big|_{-L}^{L}$$

$$A_0 = \frac{1}{2L}\left(L + \frac{1}{2}L^2 + \frac{1}{3}L^3\right) - \frac{1}{2L}\left(-L + \frac{1}{2}(-L)^2 + \frac{1}{3}(-L)^3\right)$$

$$A_0 = \frac{1}{2}\left(1 + \frac{1}{2}L + \frac{1}{3}L^2\right) - \frac{1}{2}\left(-1 + \frac{1}{2}L - \frac{1}{3}L^2\right)$$

$$A_0 = \frac{1}{2}\left(1 + \frac{1}{2}L + \frac{1}{3}L^2 + 1 - \frac{1}{2}L + \frac{1}{3}L^2\right)$$

$$A_0 = 1 + \frac{L^2}{3}$$

For A_n, we get

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$$A_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\left(\frac{n\pi x}{L}\right)\,dx$$

$$A_n = \frac{1}{L}\int_{-L}^{L} (1+x+x^2)\cos\left(\frac{n\pi x}{L}\right)\,dx$$

$$A_n = \frac{1}{L}\int_{-L}^{L} \cos\left(\frac{n\pi x}{L}\right)\,dx + \frac{1}{L}\int_{-L}^{L} x\cos\left(\frac{n\pi x}{L}\right)\,dx + \frac{1}{L}\int_{-L}^{L} x^2\cos\left(\frac{n\pi x}{L}\right)\,dx$$

$$A_n = \frac{1}{n\pi}\sin\left(\frac{n\pi x}{L}\right)\Big|_{-L}^{L} + \frac{1}{L}\int_{-L}^{L} x\cos\left(\frac{n\pi x}{L}\right)\,dx + \frac{1}{L}\int_{-L}^{L} x^2\cos\left(\frac{n\pi x}{L}\right)\,dx$$

Use integration by parts with $u = x$, $du = dx$, $dv = \cos(n\pi x/L)\,dx$, and $v = (L/n\pi)\sin(n\pi x/L)$.

$$A_n = \frac{1}{n\pi}\sin\left(\frac{n\pi x}{L}\right) + \frac{x}{n\pi}\sin\left(\frac{n\pi x}{L}\right)\Big|_{-L}^{L} - \frac{1}{n\pi}\int_{-L}^{L}\sin\left(\frac{n\pi x}{L}\right)\,dx + \frac{1}{L}\int_{-L}^{L} x^2\cos\left(\frac{n\pi x}{L}\right)\,dx$$

$$A_n = \frac{1}{n\pi}\sin\left(\frac{n\pi x}{L}\right) + \frac{x}{n\pi}\sin\left(\frac{n\pi x}{L}\right) + \frac{L}{(n\pi)^2}\cos\left(\frac{n\pi x}{L}\right)\Big|_{-L}^{L} + \frac{1}{L}\int_{-L}^{L} x^2\cos\left(\frac{n\pi x}{L}\right)\,dx$$

Use integration by parts with $u = x^2$, $du = 2x\,dx$, $dv = \cos(n\pi x/L)\,dx$, and $v = (L/n\pi)\sin(n\pi x/L)$.

$$A_n = \frac{1}{n\pi}\sin\left(\frac{n\pi x}{L}\right) + \frac{x}{n\pi}\sin\left(\frac{n\pi x}{L}\right) + \frac{L}{(n\pi)^2}\cos\left(\frac{n\pi x}{L}\right)$$

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$$+\frac{x^2}{n\pi}\sin\left(\frac{n\pi x}{L}\right)\Bigg|_{-L}^{L}-\frac{2}{n\pi}\int_{-L}^{L}x\sin\left(\frac{n\pi x}{L}\right)\,dx$$

Use integration by parts with $u=x$, $du=dx$, $dv=\sin(n\pi x/L)\,dx$, and $v=-(L/n\pi)\cos(n\pi x/L)$.

$$A_n=\frac{1}{n\pi}\sin\left(\frac{n\pi x}{L}\right)+\frac{x}{n\pi}\sin\left(\frac{n\pi x}{L}\right)+\frac{L}{(n\pi)^2}\cos\left(\frac{n\pi x}{L}\right)+\frac{x^2}{n\pi}\sin\left(\frac{n\pi x}{L}\right)$$

$$+\frac{2Lx}{(n\pi)^2}\cos\left(\frac{n\pi x}{L}\right)\Bigg|_{-L}^{L}-\frac{2L}{(n\pi)^2}\int_{-L}^{L}\cos\left(\frac{n\pi x}{L}\right)\,dx$$

$$A_n=\frac{1}{n\pi}\sin\left(\frac{n\pi x}{L}\right)+\frac{x}{n\pi}\sin\left(\frac{n\pi x}{L}\right)+\frac{L}{(n\pi)^2}\cos\left(\frac{n\pi x}{L}\right)+\frac{x^2}{n\pi}\sin\left(\frac{n\pi x}{L}\right)$$

$$+\frac{2Lx}{(n\pi)^2}\cos\left(\frac{n\pi x}{L}\right)-\frac{2L^2}{(n\pi)^3}\sin\left(\frac{n\pi x}{L}\right)\Bigg|_{-L}^{L}$$

$$A_n=\left(\frac{1+x+x^2}{n\pi}-\frac{2L^2}{(n\pi)^3}\right)\sin\left(\frac{n\pi x}{L}\right)+\left(\frac{L+2Lx}{(n\pi)^2}\right)\cos\left(\frac{n\pi x}{L}\right)\Bigg|_{-L}^{L}$$

$$A_n=\left(\frac{1+L+L^2}{n\pi}-\frac{2L^2}{(n\pi)^3}\right)\sin(n\pi)+\left(\frac{L+2L^2}{(n\pi)^2}\right)\cos(n\pi)$$

$$-\left(\frac{1-L+L^2}{n\pi}-\frac{2L^2}{(n\pi)^3}\right)\sin(-n\pi)-\left(\frac{L-2L^2}{(n\pi)^2}\right)\cos(-n\pi)$$

Using the even-odd trigonometric identities $\sin(-x)=-\sin x$ and $\cos(-x)=\cos x$, we can simplify A_n to

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$$A_n = \left(\frac{1+L+L^2}{n\pi} - \frac{2L^2}{(n\pi)^3}\right)\sin(n\pi) + \left(\frac{L+2L^2}{(n\pi)^2}\right)\cos(n\pi)$$

$$+\left(\frac{1-L+L^2}{n\pi} - \frac{2L^2}{(n\pi)^3}\right)\sin(n\pi) - \left(\frac{L-2L^2}{(n\pi)^2}\right)\cos(n\pi)$$

For $n = 1, 2, 3, \ldots$, $\sin(n\pi) = 0$, and $\cos(n\pi) = (-1)^n$, so the expression for A_n simplifies to

$$A_n = \left(\frac{L+2L^2}{(n\pi)^2}\right)(-1)^n - \left(\frac{L-2L^2}{(n\pi)^2}\right)(-1)^n$$

$$A_n = \frac{L(-1)^n + 2L^2(-1)^n}{(n\pi)^2} - \frac{L(-1)^n - 2L^2(-1)^n}{(n\pi)^2}$$

$$A_n = \frac{L(-1)^n + 2L^2(-1)^n - L(-1)^n + 2L^2(-1)^n}{(n\pi)^2}$$

$$A_n = \frac{4L^2(-1)^n}{(n\pi)^2}$$

$$A_n = \left(\frac{2L}{n\pi}\right)^2(-1)^n$$

And for B_n, we get

$$B_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$B_n = \frac{1}{L}\int_{-L}^{L} (1+x+x^2)\sin\left(\frac{n\pi x}{L}\right)\,dx$$

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$$B_n = \frac{1}{L}\int_{-L}^{L} \sin\left(\frac{n\pi x}{L}\right)\,dx + \frac{1}{L}\int_{-L}^{L} x\sin\left(\frac{n\pi x}{L}\right)\,dx + \frac{1}{L}\int_{-L}^{L} x^2\sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$B_n = -\frac{1}{n\pi}\cos\left(\frac{n\pi x}{L}\right)\Big|_{-L}^{L} + \frac{1}{L}\int_{-L}^{L} x\sin\left(\frac{n\pi x}{L}\right)\,dx + \frac{1}{L}\int_{-L}^{L} x^2\sin\left(\frac{n\pi x}{L}\right)\,dx$$

Use integration by parts with $u = x$, $du = dx$, $dv = \sin(n\pi x/L)\,dx$, and $v = -(L/n\pi)\cos(n\pi x/L)$.

$$B_n = -\frac{1}{n\pi}\cos\left(\frac{n\pi x}{L}\right) - \frac{x}{n\pi}\cos\left(\frac{n\pi x}{L}\right)\Big|_{-L}^{L} + \frac{1}{n\pi}\int_{-L}^{L}\cos\left(\frac{n\pi x}{L}\right)\,dx$$

$$+\frac{1}{L}\int_{-L}^{L} x^2\sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$B_n = -\frac{1}{n\pi}\cos\left(\frac{n\pi x}{L}\right) - \frac{x}{n\pi}\cos\left(\frac{n\pi x}{L}\right) + \frac{L}{(n\pi)^2}\sin\left(\frac{n\pi x}{L}\right)\Big|_{-L}^{L}$$

$$+\frac{1}{L}\int_{-L}^{L} x^2\sin\left(\frac{n\pi x}{L}\right)\,dx$$

Use integration by parts with $u = x^2$, $du = 2x\,dx$, $dv = \sin(n\pi x/L)\,dx$, and $v = -(L/n\pi)\cos(n\pi x/L)$.

$$B_n = -\frac{1}{n\pi}\cos\left(\frac{n\pi x}{L}\right) - \frac{x}{n\pi}\cos\left(\frac{n\pi x}{L}\right) + \frac{L}{(n\pi)^2}\sin\left(\frac{n\pi x}{L}\right)$$

$$-\frac{x^2}{n\pi}\cos\left(\frac{n\pi x}{L}\right)\Big|_{-L}^{L} + \frac{2}{n\pi}\int_{-L}^{L} x\cos\left(\frac{n\pi x}{L}\right)\,dx$$

Use integration by parts with $u = x$, $du = dx$, $dv = \cos(n\pi x/L)\,dx$, and $v = (L/n\pi)\sin(n\pi x/L)$.

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$$B_n = -\frac{1}{n\pi}\cos\left(\frac{n\pi x}{L}\right) - \frac{x}{n\pi}\cos\left(\frac{n\pi x}{L}\right) + \frac{L}{(n\pi)^2}\sin\left(\frac{n\pi x}{L}\right)$$

$$-\frac{x^2}{n\pi}\cos\left(\frac{n\pi x}{L}\right) + \frac{2Lx}{(n\pi)^2}\sin\left(\frac{n\pi x}{L}\right)\Bigg|_{-L}^{L} - \frac{2L}{(n\pi)^2}\int_{-L}^{L}\sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$B_n = -\frac{1}{n\pi}\cos\left(\frac{n\pi x}{L}\right) - \frac{x}{n\pi}\cos\left(\frac{n\pi x}{L}\right) + \frac{L}{(n\pi)^2}\sin\left(\frac{n\pi x}{L}\right)$$

$$-\frac{x^2}{n\pi}\cos\left(\frac{n\pi x}{L}\right) + \frac{2Lx}{(n\pi)^2}\sin\left(\frac{n\pi x}{L}\right) + \frac{2L^2}{(n\pi)^3}\cos\left(\frac{n\pi x}{L}\right)\Bigg|_{-L}^{L}$$

$$B_n = \left(-\frac{1+x+x^2}{n\pi} + \frac{2L^2}{(n\pi)^3}\right)\cos\left(\frac{n\pi x}{L}\right) + \left(\frac{L+2Lx}{(n\pi)^2}\right)\sin\left(\frac{n\pi x}{L}\right)\Bigg|_{-L}^{L}$$

$$B_n = \left(-\frac{1+L+L^2}{n\pi} + \frac{2L^2}{(n\pi)^3}\right)\cos(n\pi) + \left(\frac{L+2L^2}{(n\pi)^2}\right)\sin(n\pi)$$

$$-\left(-\frac{1-L+L^2}{n\pi} + \frac{2L^2}{(n\pi)^3}\right)\cos(-n\pi) - \left(\frac{L-2L^2}{(n\pi)^2}\right)\sin(-n\pi)$$

Using the even-odd trigonometric identities $\sin(-x) = -\sin x$ and $\cos(-x) = \cos x$, we can simplify B_n to

$$B_n = \left(-\frac{1+L+L^2}{n\pi} + \frac{2L^2}{(n\pi)^3}\right)\cos(n\pi) + \left(\frac{L+2L^2}{(n\pi)^2}\right)\sin(n\pi)$$

$$-\left(-\frac{1-L+L^2}{n\pi} + \frac{2L^2}{(n\pi)^3}\right)\cos(n\pi) + \left(\frac{L-2L^2}{(n\pi)^2}\right)\sin(n\pi)$$

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For $n = 1, 2, 3, \ldots$, $\sin(n\pi) = 0$, and $\cos(n\pi) = (-1)^n$, so the expression for B_n simplifies to

$$B_n = \left(-\frac{1+L+L^2}{n\pi} + \frac{2L^2}{(n\pi)^3}\right)(-1)^n - \left(-\frac{1-L+L^2}{n\pi} + \frac{2L^2}{(n\pi)^3}\right)(-1)^n$$

$$B_n = -\frac{1+L+L^2}{n\pi}(-1)^n + \frac{2L^2}{(n\pi)^3}(-1)^n + \frac{1-L+L^2}{n\pi}(-1)^n - \frac{2L^2}{(n\pi)^3}(-1)^n$$

$$B_n = \frac{1-L+L^2-1-L-L^2}{n\pi}(-1)^n$$

$$B_n = \frac{-2L}{n\pi}(-1)^n$$

$$B_n = \frac{2L}{n\pi}(-1)^{n+1}$$

Then the function's Fourier series representation on $-L \le x \le L$ is

$$f(x) = A_0 + \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right)$$

$$f(x) = 1 + \frac{L^2}{3} + \sum_{n=1}^{\infty}\left(\frac{2L}{n\pi}\right)^2(-1)^n\cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty}\frac{2L}{n\pi}(-1)^{n+1}\sin\left(\frac{n\pi x}{L}\right)$$

$$f(x) = 1 + \frac{L^2}{3} + \frac{4L^2}{\pi^2}\sum_{n=1}^{\infty}\frac{(-1)^n}{n^2}\cos\left(\frac{n\pi x}{L}\right) + \frac{2L}{\pi}\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n}\sin\left(\frac{n\pi x}{L}\right)$$

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Topic: Fourier series representations

Question: Find the Fourier series representation of the function on $-L \leq x \leq L$.

$$f(x) = x - x^3$$

Answer choices:

A $$f(x) = \frac{L}{\pi} + \frac{2L}{\pi} \sum_{n=1}^{\infty} \left(\frac{L^2 - 1}{n} - \frac{6L^2}{n^3 \pi^2} \right) (-1)^n \cos \left(\frac{n \pi x}{L} \right)$$

B $$f(x) = \frac{L}{\pi} + \sum_{n=1}^{\infty} \left(\frac{L^2 - 1}{n} - \frac{6L^2}{n^3 \pi^2} \right) \cos \left(\frac{n \pi x}{L} \right)$$

C $$f(x) = \frac{2L}{\pi} \sum_{n=1}^{\infty} \left(\frac{L^2 - 1}{n} - \frac{6L^2}{n^3 \pi^2} \right) (-1)^n \sin \left(\frac{n \pi x}{L} \right)$$

D $$f(x) = \sum_{n=1}^{\infty} \left(\frac{L^2 - 1}{n} - \frac{6L^2}{n^3 \pi^2} \right) \sin \left(\frac{n \pi x}{L} \right)$$

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Solution: C

The function is odd because $f(-x) = -f(x)$,

$$f(-x) = -x - (-x)^3$$

$$f(-x) = -x + x^3$$

$$f(-x) = -(x - x^3)$$

$$f(-x) = -f(x)$$

Because the function is odd, $A_0 = 0$ and $A_n = 0$, and we only need to calculate B_n. For B_n, we get

$$B_n = \frac{2}{L}\int_0^L f(x)\sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$B_n = \frac{2}{L}\int_0^L (x - x^3)\sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$B_n = \frac{2}{L}\int_0^L x\sin\left(\frac{n\pi x}{L}\right)\,dx - \frac{2}{L}\int_0^L x^3\sin\left(\frac{n\pi x}{L}\right)\,dx$$

Use integration by parts with $u = x^3$, $du = 3x^2\,dx$, $dv = \sin(n\pi x/L)\,dx$, and $v = -(L/n\pi)\cos(n\pi x/L)$.

$$B_n = \frac{2x^3}{n\pi}\cos\left(\frac{n\pi x}{L}\right)\Big|_0^L + \frac{2}{L}\int_0^L x\sin\left(\frac{n\pi x}{L}\right)\,dx - \frac{6}{n\pi}\int_0^L x^2\cos\left(\frac{n\pi x}{L}\right)\,dx$$

Use integration by parts with $u = x^2$, $du = 2x\,dx$, $dv = \cos(n\pi x/L)\,dx$, and $v = (L/n\pi)\sin(n\pi x/L)$.

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$$B_n = \frac{2x^3}{n\pi}\cos\left(\frac{n\pi x}{L}\right) - \frac{6Lx^2}{(n\pi)^2}\sin\left(\frac{n\pi x}{L}\right)\Bigg|_0^L + \left(\frac{2}{L} + \frac{12L}{(n\pi)^2}\right)\int_0^L x\sin\left(\frac{n\pi x}{L}\right)\,dx$$

Use integration by parts with $u = x$, $du = dx$, $dv = \sin(n\pi x/L)\,dx$, and $v = -(L/n\pi)\cos(n\pi x/L)$.

$$B_n = \frac{2x^3}{n\pi}\cos\left(\frac{n\pi x}{L}\right) - \frac{6Lx^2}{(n\pi)^2}\sin\left(\frac{n\pi x}{L}\right)$$

$$-\left(\frac{2}{L} + \frac{12L}{(n\pi)^2}\right)\frac{Lx}{n\pi}\cos\left(\frac{n\pi x}{L}\right)\Bigg|_0^L + \left(\frac{2}{L} + \frac{12L}{(n\pi)^2}\right)\frac{L}{n\pi}\int_0^L \cos\left(\frac{n\pi x}{L}\right)\,dx$$

$$B_n = \frac{2x^3}{n\pi}\cos\left(\frac{n\pi x}{L}\right) - \frac{6Lx^2}{(n\pi)^2}\sin\left(\frac{n\pi x}{L}\right)$$

$$-\left(\frac{2}{L} + \frac{12L}{(n\pi)^2}\right)\frac{Lx}{n\pi}\cos\left(\frac{n\pi x}{L}\right) + \left(\frac{2}{L} + \frac{12L}{(n\pi)^2}\right)\frac{L^2}{(n\pi)^2}\sin\left(\frac{n\pi x}{L}\right)\Bigg|_0^L$$

$$B_n = \left(\frac{2x^3 - 2x}{n\pi} - \frac{12L^2x}{(n\pi)^3}\right)\cos\left(\frac{n\pi x}{L}\right) + \left(\frac{2L - 6Lx^2}{(n\pi)^2} + \frac{12L^3}{(n\pi)^4}\right)\sin\left(\frac{n\pi x}{L}\right)\Bigg|_0^L$$

$$B_n = \left(\frac{2L^3 - 2L}{n\pi} - \frac{12L^3}{(n\pi)^3}\right)\cos(n\pi) + \left(\frac{2L - 6L^3}{(n\pi)^2} + \frac{12L^3}{(n\pi)^4}\right)\sin(n\pi) - \frac{12L^3}{(n\pi)^4}\sin(0)$$

For $n = 1, 2, 3, \ldots$, $\sin(n\pi) = 0$, and $\cos(n\pi) = (-1)^n$, so the expression for B_n simplifies to

$$B_n = \left(\frac{2L^3 - 2L}{n\pi} - \frac{12L^3}{(n\pi)^3}\right)(-1)^n$$

Then the function's Fourier series representation on $-L \le x \le L$ is

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$$f(x) = A_0 + \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right)$$

$$f(x) = \frac{2L}{\pi} \sum_{n=1}^{\infty} \left(\frac{L^2 - 1}{n} - \frac{6L^2}{n^3\pi^2}\right)(-1)^n \sin\left(\frac{n\pi x}{L}\right)$$

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Topic: Fourier series representations

Question: Find the Fourier series representation of the function on $-L \leq x \leq L$.

$$f(x) = x^2 + x^4$$

Answer choices:

A $$f(x) = \sum_{n=1}^{\infty} \left(\frac{2L^2 + 1}{n^2} - \frac{12L^2}{n^4 \pi^2} \right) (-1)^n \sin\left(\frac{n\pi x}{L} \right)$$

B $$f(x) = \left(\frac{2L}{\pi} \right)^2 \sum_{n=1}^{\infty} \left(\frac{2L^2 + 1}{n^2} - \frac{12L^2}{n^4 \pi^2} \right) (-1)^n \sin\left(\frac{n\pi x}{L} \right)$$

C $$f(x) = \frac{L^2 + L^4}{5} + \sum_{n=1}^{\infty} \left(\frac{2L^2 + 1}{n^2} - \frac{12L^2}{n^4 \pi^2} \right) (-1)^n \cos\left(\frac{n\pi x}{L} \right)$$

D $$f(x) = \frac{L^2}{3} + \frac{L^4}{5} + \left(\frac{2L}{\pi} \right)^2 \sum_{n=1}^{\infty} \left(\frac{2L^2 + 1}{n^2} - \frac{12L^2}{n^4 \pi^2} \right) (-1)^n \cos\left(\frac{n\pi x}{L} \right)$$

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Solution: D

The function is even because $f(-x) = f(x)$,

$$f(-x) = (-x)^2 + (-x)^4$$

$$f(-x) = x^2 + x^4$$

$$f(-x) = f(x)$$

Because the function is even, $B_n = 0$ and we only need to calculate A_0 and A_n. For A_0, we get

$$A_0 = \frac{1}{L}\int_0^L f(x)\ dx$$

$$A_0 = \frac{1}{L}\int_0^L x^2 + x^4\ dx$$

$$A_0 = \frac{1}{L}\left(\frac{1}{3}x^3 + \frac{1}{5}x^5\right)\Bigg|_0^L$$

$$A_0 = \frac{1}{L}\left(\frac{1}{3}L^3 + \frac{1}{5}L^5\right)$$

$$A_0 = \frac{L^2}{3} + \frac{L^4}{5}$$

For A_n, we get

$$A_n = \frac{2}{L}\int_0^L f(x)\cos\left(\frac{n\pi x}{L}\right)\ dx$$

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$$A_n = \frac{2}{L}\int_0^L (x^2 + x^4)\cos\left(\frac{n\pi x}{L}\right)\,dx$$

$$A_n = \frac{2}{L}\int_0^L x^2\cos\left(\frac{n\pi x}{L}\right)\,dx + \frac{2}{L}\int_0^L x^4\cos\left(\frac{n\pi x}{L}\right)\,dx$$

Use integration by parts with $u = x^4$, $du = 4x^3\,dx$, $dv = \cos(n\pi x/L)\,dx$, and $v = (L/n\pi)\sin(n\pi x/L)$.

$$A_n = \frac{2x^4}{n\pi}\sin\left(\frac{n\pi x}{L}\right)\Big|_0^L + \frac{2}{L}\int_0^L x^2\cos\left(\frac{n\pi x}{L}\right)\,dx - \frac{8}{n\pi}\int_0^L x^3\sin\left(\frac{n\pi x}{L}\right)\,dx$$

Use integration by parts with $u = x^3$, $du = 3x^2\,dx$, $dv = \sin(n\pi x/L)\,dx$, and $v = -(L/n\pi)\cos(n\pi x/L)$.

$$A_n = \frac{2x^4}{n\pi}\sin\left(\frac{n\pi x}{L}\right) + \frac{8Lx^3}{(n\pi)^2}\cos\left(\frac{n\pi x}{L}\right)\Big|_0^L + \frac{2}{L}\int_0^L x^2\cos\left(\frac{n\pi x}{L}\right)\,dx - \frac{24L}{(n\pi)^2}\int_0^L x^2\cos\left(\frac{n\pi x}{L}\right)\,dx$$

$$A_n = \frac{2x^4}{n\pi}\sin\left(\frac{n\pi x}{L}\right) + \frac{8Lx^3}{(n\pi)^2}\cos\left(\frac{n\pi x}{L}\right)\Big|_0^L + \left(\frac{2}{L} - \frac{24L}{(n\pi)^2}\right)\int_0^L x^2\cos\left(\frac{n\pi x}{L}\right)\,dx$$

Use integration by parts with $u = x^2$, $du = 2x\,dx$, $dv = \cos(n\pi x/L)\,dx$, and $v = (L/n\pi)\sin(n\pi x/L)$.

$$A_n = \frac{2x^4}{n\pi}\sin\left(\frac{n\pi x}{L}\right) + \frac{8Lx^3}{(n\pi)^2}\cos\left(\frac{n\pi x}{L}\right) + \left(\frac{2x^2}{n\pi} - \frac{24L^2x^2}{(n\pi)^3}\right)\sin\left(\frac{n\pi x}{L}\right)\Big|_0^L$$

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$$-\left(\frac{4}{n\pi}-\frac{48L^2}{(n\pi)^3}\right)\int_0^L x\sin\left(\frac{n\pi x}{L}\right)\,dx$$

Use integration by parts with $u = x$, $du = dx$, $dv = \sin(n\pi x/L)\,dx$, and $v = -(L/n\pi)\cos(n\pi x/L)$.

$$A_n=\frac{2x^4}{n\pi}\sin\left(\frac{n\pi x}{L}\right)+\frac{8Lx^3}{(n\pi)^2}\cos\left(\frac{n\pi x}{L}\right)+\left(\frac{2x^2}{n\pi}-\frac{24L^2x^2}{(n\pi)^3}\right)\sin\left(\frac{n\pi x}{L}\right)$$

$$+\left(\frac{4Lx}{(n\pi)^2}-\frac{48L^3x}{(n\pi)^4}\right)\cos\left(\frac{n\pi x}{L}\right)\Bigg|_0^L-\left(\frac{4L}{(n\pi)^2}-\frac{48L^3}{(n\pi)^4}\right)\int_0^L\cos\left(\frac{n\pi x}{L}\right)\,dx$$

$$A_n=\frac{2x^4}{n\pi}\sin\left(\frac{n\pi x}{L}\right)+\frac{8Lx^3}{(n\pi)^2}\cos\left(\frac{n\pi x}{L}\right)+\left(\frac{2x^2}{n\pi}-\frac{24L^2x^2}{(n\pi)^3}\right)\sin\left(\frac{n\pi x}{L}\right)$$

$$+\left(\frac{4Lx}{(n\pi)^2}-\frac{48L^3x}{(n\pi)^4}\right)\cos\left(\frac{n\pi x}{L}\right)-\left(\frac{4L^2}{(n\pi)^3}-\frac{48L^4}{(n\pi)^5}\right)\sin\left(\frac{n\pi x}{L}\right)\Bigg|_0^L$$

$$A_n=\left(\frac{2x^4+2x^2}{n\pi}-\frac{24L^2x^2+4L^2}{(n\pi)^3}+\frac{48L^4}{(n\pi)^5}\right)\sin\left(\frac{n\pi x}{L}\right)$$

$$+\left(\frac{8Lx^3+4Lx}{(n\pi)^2}-\frac{48L^3x}{(n\pi)^4}\right)\cos\left(\frac{n\pi x}{L}\right)\Bigg|_0^L$$

For $n = 1, 2, 3, \ldots$, $\sin(n\pi) = 0$, and $\cos(n\pi) = (-1)^n$, so the expression for A_n simplifies to

$$A_n=\left(\frac{8Lx^3+4Lx}{(n\pi)^2}-\frac{48L^3x}{(n\pi)^4}\right)(-1)^n\Bigg|_0^L$$

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$$A_n = \left(\frac{8L^4 + 4L^2}{(n\pi)^2} - \frac{48L^4}{(n\pi)^4} \right)(-1)^n$$

$$A_n = \left(\frac{2L}{\pi} \right)^2 \left(\frac{2L^2 + 1}{n^2} - \frac{12L^2}{n^4\pi^2} \right)(-1)^n$$

Then the function's Fourier series representation on $-L \leq x \leq L$ is

$$f(x) = A_0 + \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi x}{L} \right) + \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L} \right)$$

$$f(x) = \frac{L^2}{3} + \frac{L^4}{5} + \left(\frac{2L}{\pi} \right)^2 \sum_{n=1}^{\infty} \left(\frac{2L^2 + 1}{n^2} - \frac{12L^2}{n^4\pi^2} \right)(-1)^n \cos\left(\frac{n\pi x}{L} \right)$$

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